

1. Administrative & Study Identification

Full Study Title: A Study of Atezolizumab Plus Tiragolumab and Atezolizumab Plus Placebo as First-Line Treatment in Participants With Recurrent/Metastatic PD-L1 Positive Squamous Cell Carcinoma of the Head and Neck **Short Title/Acronym:** SKYSCRAPER-09 **Protocol Number:** B042533 **NCT Number:** NCT04665843 **Posting ID:** 336242404924023701 **Sponsor/Company Name:** F. Hoffmann-La Roche Ltd (Roche) **Investigational Product:** Atezolizumab (Trade Name: Tecentriq; ATC Code: L01FF05; anti-PD-L1) and Tiragolumab (anti-TIGIT) **Phase of Development:** Phase 2 **Trial Status:** COMPLETED (Study has been completed) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of atezolizumab plus tiragolumab and atezolizumab plus placebo as first-line (1L) treatment in recurrent/metastatic PD-L1-positive squamous cell carcinoma of the head and neck (SCCHN) on the basis of confirmed objective response rate (ORR). **Secondary Objectives:** To evaluate progression-free survival (PFS), overall survival (OS), duration of response (DOR), safety, pharmacokinetics, and immunogenicity of atezolizumab and tiragolumab.

2.2 Background & Rationale To address the medical necessity for improved first-line treatment options in patients with recurrent or metastatic PD-L1-positive SCCHN. While current standard of care includes PD-1/PD-L1 inhibitors, many patients do not achieve durable responses. Combining the anti-TIGIT monoclonal antibody tiragolumab with the PD-L1 inhibitor atezolizumab aims to synergistically restore tumor-specific T-cell functionality and overcome resistance to PD-L1 blockade, potentially improving response rates and survival outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 2:1 (Atezolizumab + Tiragolumab vs. Atezolizumab + Placebo)

3.2 Planned Enrollment & Duration Targeted Enrollment: 123 **Number of Sites:** Global, multicenter **Estimated Study Start:** December 2020 **Estimated Primary Completion:** 20-Sep-2023 (Study Completion: 27-Aug-2025)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Histologically or cytologically confirmed recurrent/metastatic SCCHN involving the oropharynx, oral cavity, larynx, or hypopharynx, that is considered incurable by local therapies. 3. Known results from human papillomavirus (HPV) status test for oropharyngeal carcinoma. 4. No prior systemic therapy for metastatic and/or recurrent SCCHN. 5. Measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. 6. Tumor PD-L1 expression as determined by PD-L1 immunohistochemistry assay. 7. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 8. Life expectancy $\geq$ 12 weeks.

4.2 Exclusion Criteria 1. Disease suitable for local therapy with curative intent. 2. Progressive or recurrent disease within 6 months of the last dose of curative intent systemic treatment for locally advanced SCCHN. 3. Rapidly progressing disease in the opinion of the treating investigator. 4. Grade $\geq$ 2 unresolved toxicity related to surgery or other prior therapies. 5. Symptomatic, untreated, or actively progressing central nervous system (CNS) metastases. 6. History of leptomeningeal disease. 7. Active or history of autoimmune disease or immune deficiency. 8. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or idiopathic pneumonitis, or evidence of active pneumonitis on screening chest computed tomography (CT) scan. 9. History of additional malignancy other than SCCHN within 5 years prior to randomization. 10. Prior treatment with CD137 agonists or immune checkpoint blockade therapies, including anti-CTLA-4, anti-TIGIT, anti-PD-L1, and anti-PD-1 therapeutic antibodies. 11. Treatment with systemic immunostimulatory agents or systemic immunosuppressive medication. 12. Pregnancy or breastfeeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Experimental Arm: Tiragolumab 600 mg IV + Atezolizumab 1200 mg IV every 3 weeks (Q3W). - Control Arm: Atezolizumab 1200 mg IV Q3W + Placebo IV Q3W. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Until disease progression, loss of clinical benefit, or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for SCCHN symptoms, palliative radiation (with specific restrictions). **Prohibited:** Concurrent administration of other investigational agents, systemic immunosuppressive medications, live vaccines, and specific immunostimulatory agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Tumor biopsy for PD-L1 testing, Baseline Imaging (RECIST v1.1), HPV status testing
Baseline	Day 0 Randomization, First IV Infusion (Atezolizumab + Tiragolumab/Placebo)	Interim	Every 3 Weeks IV Infusion, Safety Labs, Efficacy Assessment, AE Monitoring
End of Study	Variable End of Treatment Visit, Final Vital Signs, Survival Follow-up (Every 3 months post-treatment)		

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Confirmed Objective Response Rate (ORR)

7.2 Secondary Outcomes 1. Duration of Response (DOR) 2. Progression-Free Survival (PFS) 3. Overall Survival (OS) 4. Incidence and severity of Adverse Events (Safety profile) 5. Pharmacokinetics and Immunogenicity

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through patient assessment and laboratory evaluations every 3 weeks prior to each infusion cycle. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, with specific guidelines for immune-mediated adverse events (imAEs). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned by the sponsor to oversee safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Designed and powered to detect a statistically significant difference in ORR between the combination therapy arm and the control arm. **Handling of Missing Data:** Censoring applied for survival endpoints (OS/PFS) at the last known date the patient was alive and progression-free.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to assure protocol adherence, safety reporting, and data integrity. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with complete

audit trails, source data verification, and data clarification forms managed by Hoffmann-La Roche.

11. Study Identifiers & Governance

Primary ID: B042533 **Posting ID:** 336242404924023701 **Registry Number:** NCT04665843 **Sponsor/Lead Organization:** F. Hoffmann-La Roche Ltd (Roche) **Study Title:** SKYSCRAPER-09 **Official Regulatory Title:** A Phase II, Randomized, Double Blind Study of Atezolizumab Plus Tiragolumab and Atezolizumab Plus Placebo as First-Line Treatment in Patients With Recurrent/Metastatic PD-L1 Positive Squamous Cell Carcinoma of the Head and Neck

12. Clinical Framework (Oncology Specific)

Primary Condition: Squamous Cell Carcinoma of Head and Neck (Preferred UMLS Name: Head and Neck Squamous Cell Carcinoma) **Diagnosis Threshold:** PD-L1 Positive **Medical Methodology:** Immune Checkpoint Inhibition (anti-PD-L1 and anti-TIGIT blockade) **Optimization Target:** Objective tumor response, prolonged progression-free survival, and overall survival

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for systemic IV immunotherapy administration. **Education Protocol (HCP):** Training on immune-mediated adverse event identification and management. **Education Protocol (Patient):** Patient consent and education regarding potential immune-related side effects and infusion schedule. **Quality Audit Mechanism:** Centralized RECIST v1.1 imaging review and PD-L1 central laboratory testing validation.

14. Observation & Follow-Up Schedule

Total Duration: Treatment until progression or unacceptable toxicity; survival follow-up up to 43 months depending on patient response. **Onsite Visit Cadence:** Every 3 weeks (Q3W) for treatment infusions and assessments. **Remote Monitoring Frequency:** Telephone follow-ups every 3 months during the post-treatment survival phase. **Checklist Review Interval:** Routine safety lab checks and imaging scans periodically per protocol (typically every 6-9 weeks for RECIST assessment).

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					